

Echoes and Delays: Time-to-Build in Production Networks

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Motivation

- Firms in a modern economy rely on a **complex network of suppliers**, whose **time-to-build** varies significantly
 - ▶ Ex.: < 1 month for furniture assembly, 1-2 years for container ships
- Most of the literature on production networks ignores **time-to-build** and **delays**
 - ▶ Acemoglu et al. (2012), Baqaee and Farhi (2019, 2020), ... essentially static
 - ▶ Roundabout production: inputs are produced and used in the same period, so disruptions are resolved **within the period**
- How does the introduction of **time-to-build** or **delivery lags** affect the dynamics of production networks?

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- We propose a simple model to introduce **time-to-build** (T2B) in production networks
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- We analyze how T2B contributes to the propagation of shocks:
 1. **Persistence, delays and bottlenecks** $\Rightarrow$ shocks take much longer to work through the economy
 2. **Echoes and endogenous fluctuations**
 $\Rightarrow$ cycles in the network make shocks come back, creating cyclical fluctuations
 3. **Dynamic sectoral comovements and aggregation**
 $\Rightarrow$ delays damp aggregate fluctuations at business-cycle frequencies

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 3. **Dynamic sectoral comovements** and **aggregation**
 $\Rightarrow$ delays damp aggregate fluctuations at business-cycle frequencies
- We test the model's prediction in monthly U.S. industrial production data
 - ▶ customers move with their suppliers at the measured delays
 - ▶ aggregate implications visible in the data

Related Literature

- Shock propagation in production networks
 - ▶ Acemoglu et al. (2012), Barrot and Sauvagnat (2016), Acemoglu et al. (2017), Baqaee and Farhi (2019), Carvalho et al. (2021), Ghassibe (2024), ...
- Time-to-build
 - ▶ Kydland and Prescott (1982), Schwartzman (2014), ...
 - ▶ In production networks: Long and Plosser (1983), Carvalho and Reischer (2021), Liu and Tsyvinski (2024), Bizzarri (2024), Bizzarri, Pangallo and Queirós (2025), Leng et al. (2025)
- Endogenous fluctuations in multi-sector economies
 - ▶ Benhabib and Nishimura (1979, 1985, ...)
- Delays in supply chains
 - ▶ Djankov et al. (2010), Hummels and Schaur (2013), Alessandria et al. (2023), Carreras-Valle and Ferrari (2025)

More

Input-output network

- BEA use table for 2017: 402 industries, 371 sectors kept

Measuring delays

$$\text{backlog ratio} = \frac{\text{stock value of unfilled orders}}{\text{flow value of goods delivered per month}}$$

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1. Census M3 survey: 7 durable-goods industries, each ratio applied to all its sectors: 124 sectors
2. Compustat order backlog: listed firms, but firm level and many sectors outside manufacturing: 135 sectors
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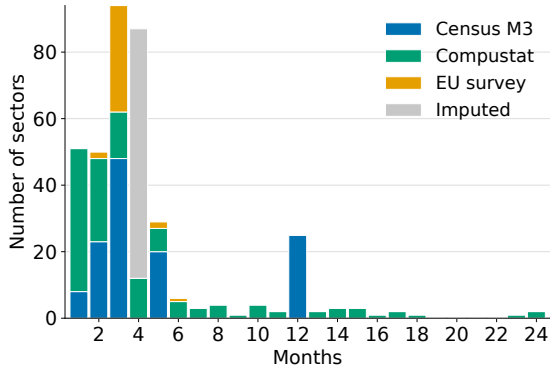
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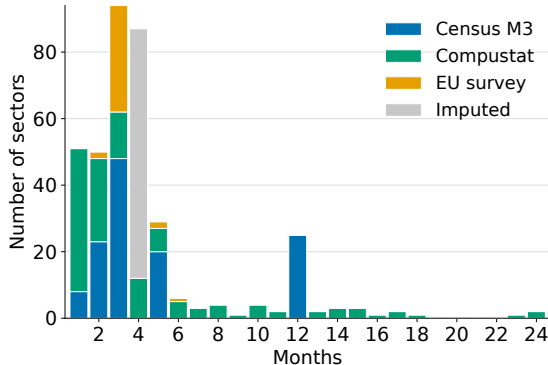
Later on, we show that this measure has bite in the data.

Backlog Distribution



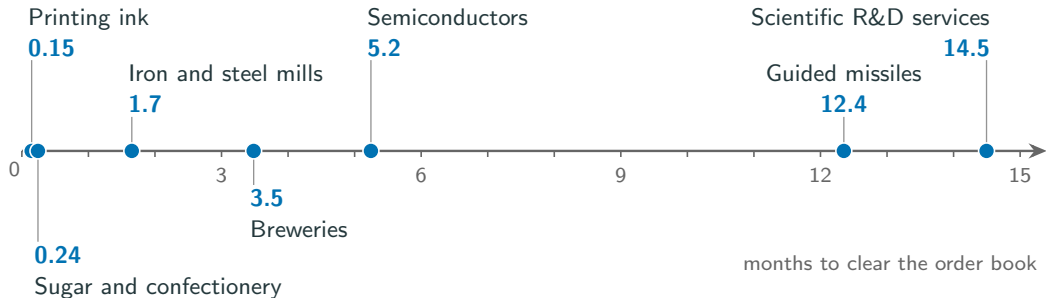
- Average 4.4 months, 5.0 weighted by value added

Backlog Distribution



- Average 4.4 months, 5.0 weighted by value added
- Measured for 296 sectors (53.5% of value added)
- The rest (gray), mostly agriculture and services, get the average of measured manufacturing, 4 months

Delays range from days to more than a year



Backlog ratios in months, before rounding

Model

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- Time is discrete
- Representative household with inelastic labor supply, discount factor β
- Sectors $n = 1, \dots, N$ with production

$$y_{nt} = A_{nt} F_n(l_{nt}, x_{n1,t}, \dots, x_{nN,t})$$

where A_{nt} is productivity, l_{nt} labor, $x_{nm,t}$ input of good m

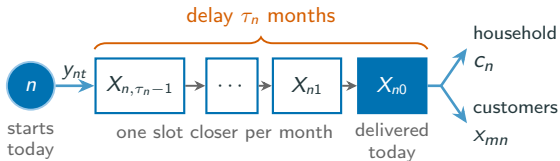
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- Time-to-build modeled as **delivery lags**:
 - ▶ Goods of sector n take τ_n periods to be delivered (the *delay* of n)
 - ▶ $X_{n\tau} \equiv$ stock of good n scheduled for delivery in τ periods
- No capital, no storage between periods



Planning Problem

Each period the planner sees the productivities $\{A_n\}$ and the goods in the pipeline $\{X_{n\tau}\}$, the **state**, and chooses consumption, labor and inputs:

$$V(\{A_n\}, \{X_{1\tau}\}_{\tau=0}^{\tau_1-1}, \dots, \{X_{N\tau}\}_{\tau=0}^{\tau_N-1}) = \max_{c_n, l_n, x_{nm}, y_n} U(c_1, \dots, c_N) \\ + \beta E[V(\{A'_n\}, \{X'_{1\tau}\}_{\tau=0}^{\tau_1-1}, \dots, \{X'_{N\tau}\}_{\tau=0}^{\tau_N-1})]$$

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$$\begin{aligned} X'_{n\tau} &= X_{n,\tau+1} \text{ for } 0 \leq \tau < \tau_n - 1 && \text{goods in the pipeline move one period closer} \\ X'_{n,\tau_n-1} &= y_n && \text{output started today is delivered in } \tau_n \text{ periods} \\ X_{n0} &\geq c_n + \sum_m x_{mn} && \text{goods delivered today are consumed or used as inputs} \\ y_n &= A_n F_n(l_n, x_{n1}, \dots, x_{nN}) \end{aligned}$$

⇒ The state is the whole pipeline, 371 sectors × their lags: intractable in general

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- Way out: **log utility** and **Cobb-Douglas** production with constant returns

$$U = \log \left(\prod_n c_n^{\gamma_n} \right), \quad F_n(l, x_1, \dots, x_N) = l^{\alpha_n} \prod_m x_m^{\omega_{nm}}, \quad \alpha_n + \sum_m \omega_{nm} = 1$$

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Proposition

Under these assumptions $l_n = \bar{l}_n$, $c_n = \bar{c}_n X_{n0}$, $x_{mn} = \bar{x}_{mn} X_{n0}$ (constants), and the value function is

$$V = \sum_{n=1}^N \sum_{\tau=0}^{\tau_n-1} \beta^\tau \zeta_n \log X_{n\tau} + G(\{A_n\}),$$

where $\zeta_n = \gamma_n + \sum_m \omega_{mn} \beta^{\tau_m} \zeta_m$ and $G(\{A_n\}) = \sum_n \beta^{\tau_n} \zeta_n \log A_n + \beta E[G(\{A'_n\})]$.

- V = **value of the goods in the pipeline**(ζ) + **value of productivity**(G)

What ζ_n and G capture

- ζ_n = value of one more percent of good n , *once delivered*:

$$\zeta_n = \underbrace{\gamma_n}_{\text{consumption of } n} + \sum_m \underbrace{\omega_{mn}}_{\substack{\text{output of customer } m \\ \text{rises by } \omega_{mn} \%}} \underbrace{\beta^{\tau_m}}_{\substack{\text{delivered} \\ \tau_m \text{ periods later}}} \underbrace{\zeta_m}_{\substack{\text{worth } \zeta_m \\ \text{once delivered}}}$$

- Iterating: ζ_n adds up every supply chain from n to the household, weighted by its input shares and discounted by the delays along the way
 - ▶ Without delays, $\zeta = (\mathbf{I} - \mathbf{\Omega}')^{-1} \gamma$: the Domar weights of the static network

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- G = value of productivity:
 - ▶ With i.i.d. shocks, $G = \sum_n \beta^{\tau_n} \zeta_n \log A_n$: a 1% shock to n is worth $\beta^{\tau_n} \zeta_n$
 - ▶ the extra output is delivered τ_n periods later (β^{τ_n}), and is then worth ζ_n
 - ▶ in the equilibrium $\beta^{\tau_n} \zeta_n$ is the *forward value of production*
- Decentralization: prices, GDP, and Hulten's theorem in the backup slides

- In log-deviation from steady state:

$$\hat{y}_{nt} = \hat{A}_{nt} + \sum_m \omega_{nm} \hat{y}_{m,t-\tau_m}$$

each input enters with its supplier's delay τ_m

- $\text{VAR}(\tau_{\max})$ representation:

$$\hat{\mathbf{y}}_t = \mathbf{\Omega}_1 \hat{\mathbf{y}}_{t-1} + \dots + \mathbf{\Omega}_{\tau_{\max}} \hat{\mathbf{y}}_{t-\tau_{\max}} + \hat{\mathbf{A}}_t, \quad \mathbf{\Omega}_{\tau} = \mathbf{\Omega} \text{diag}(\mathbf{1}\{\tau = \tau_m\})$$

- ▶ $\mathbf{\Omega}_{\tau}$ keeps the columns of $\mathbf{\Omega}$ for the suppliers with delay τ

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- ▶ Ω_τ keeps the columns of Ω for the suppliers with delay τ

- Nested cases:

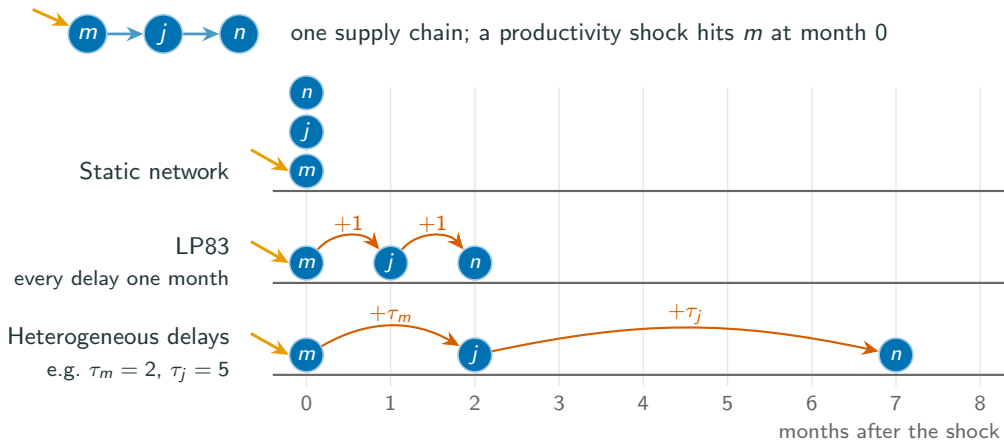
- ▶ Roundabout production (static network): no time-to-build

$$\hat{y}_t = \hat{A}_t + \Omega \hat{y}_t \Rightarrow \hat{y}_t = \hat{A}_t + \Omega \hat{A}_t + \Omega^2 \hat{A}_t + \dots = [\mathbf{I} - \Omega]^{-1} \hat{A}_t \quad (\text{Leontief inverse})$$

- ▶ Long and Plosser (1983), LP83: one-period time-to-build

$$\hat{y}_t = \hat{A}_t + \Omega \hat{y}_{t-1} \Rightarrow \hat{y}_t = \hat{A}_t + \Omega \hat{A}_{t-1} + \Omega^2 \hat{A}_{t-2} + \dots$$

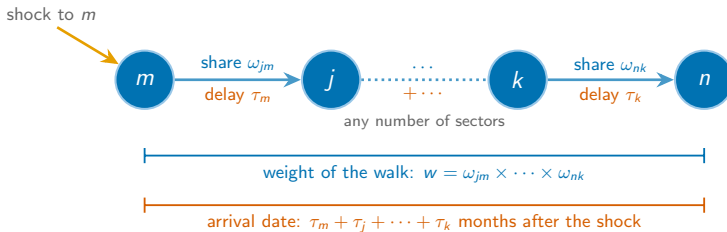
Same shock, three timings



⇒ Same chain in all three economies; the delays set only when each sector responds

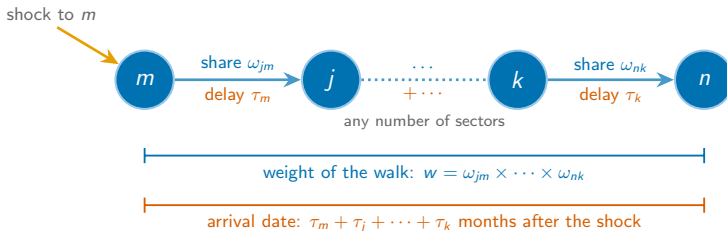
Propagation along walks

- *Walk*: a supply chain of any length, $m \rightarrow j \rightarrow \dots \rightarrow k \rightarrow n$
 - ▶ its *weight*: the product of the input shares along it
 - ▶ its *arrival date*: the sum of the delays along it, the last sector excepted



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- Response of n at date t : add up the walks that end at n and arrive at t , each weighted by its weight times the shock at its start

$$\hat{y}_{nt} = \sum_m \sum_{\substack{\varsigma: m \rightarrow n \\ \text{arriving at } t}} w(\varsigma) \hat{A}_{m, t - \tau(\varsigma)}$$

Delays change the timing of a shock, not its size

- Add up the response of n over all dates: every walk that ends at n now counts once, whatever its arrival date

$$\sum_t \hat{y}_{nt} = \sum_m \sum_{\varsigma: m \rightarrow n} w(\varsigma) \hat{A}_m = \left[(\mathbf{I} - \mathbf{\Omega})^{-1} \hat{\mathbf{A}} \right]_n$$

- This is the response of the [static network](#) on impact (Leontief inverse): the weights do not depend on the delays, only the arrival dates do
 - ▶ LP83 is the case where every delay is one period: same total, dates 1, 2, 3, ... by walk length

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 - ▶ LP83 is the case where every delay is one period: same total, dates 1, 2, 3, ... by walk length
- *Cycle*: a walk from a sector back to itself; its arrival date is the cycle's *duration*. A shock comes back every duration, scaled by the cycle's weight: an *echo*
 - ▶ simplest cycle, the *self-loop* (a sector using its own output): the shock returns after τ_n periods with weight ω_{nn} , after $2\tau_n$ with ω_{nn}^2 , ...

⇒ Delays leave the total effect of a shock unchanged and change only its timing

Persistence and Delays

Persistence: the average duration of a shock

- From now on, shocks are i.i.d.: any persistence comes from propagation
- Delays leave the total effect of a shock unchanged; they set *when* it arrives

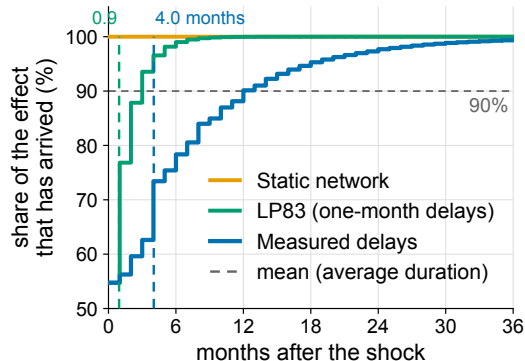
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- Delays leave the total effect of a shock unchanged; they set *when* it arrives
- We summarize the timing by one number, the **average duration** $\mathcal{T}$: the mean date at which the effect of the shock arrives

$$p_t = \frac{\hat{y}_t}{\text{CIR}}, \quad \mathcal{T} = \sum_{t \geq 0} t p_t$$

- ▶ $\hat{y}_t = \sum_n \mu_n \hat{y}_{nt}$: response of aggregate output in month t (value-added weights μ_n)
 - ▶ $\text{CIR} = \sum_{t \geq 0} \hat{y}_t$: the total effect, summed over all months
 - ▶ p_t : share of the total effect that arrives in month t
- Two shocks: a unit shock to *every* sector, and a unit shock to a *single* sector

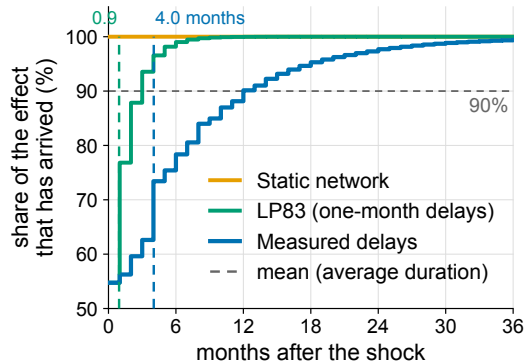
Shocks take several times longer to work through the economy



- Static network: all on impact
- LP83 and our model: 55% on impact (shocked sectors' own output)
- LP83: the rest within a few months
- Our model: a tenth still to come after 12 months

Unit shock to every sector: share of the effect arrived
by month t

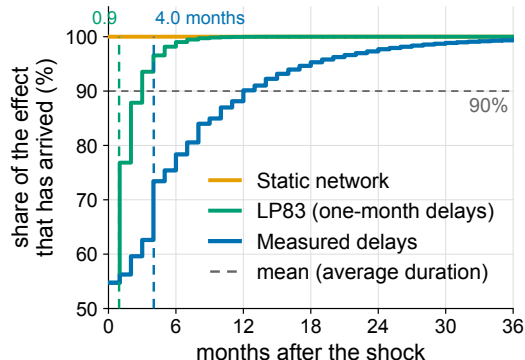
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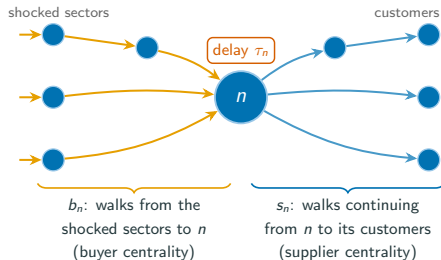
- Shocks to a single sector: a fifth of the sectors take 7 months or more (none above 3 in LP83); guided missiles, 32.7 months

Average duration = delays $\times$ traffic

- Each walk carries part of the total effect. $\mathcal{T}$ is the average arrival date across walks, each weighted by its part
- A walk is delayed τ_n each time it passes through n (the last sector excepted), so

$$\mathcal{T} = \sum_n \tau_n v_n$$

- v_n : the **traffic** through n , the average number of passages through n
- To compute it, we count the walks through n : those arriving at n , and those leaving it



Two centralities: walks into n and walks out of n

- $[\Omega^k]_{nm}$: total weight of the walks of k steps from m to n
- Buyer centrality b_n : every walk from a shocked sector to n

$$b_n = \sum_m \sum_{\varsigma: m \rightarrow n} w(\varsigma) \hat{A}_m = [(\mathbf{I} + \Omega + \Omega^2 + \dots) \hat{\mathbf{A}}]_n = [(\mathbf{I} - \Omega)^{-1} \hat{\mathbf{A}}]_n$$

- ▶ the cumulative response of n 's output to the shock
- ▶ large if n buys, directly or indirectly, from the shocked sectors

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- ▶ the cumulative response of n 's output to the shock
 - ▶ large if n buys, directly or indirectly, from the shocked sectors
- **Supplier centrality** s_n : every walk of at least one step from n to a sector j , weighted by j 's size μ_j

$$s_n = \sum_j \mu_j \sum_{\varsigma: n \rightarrow j} w(\varsigma) = [\mu'(\Omega + \Omega^2 + \dots)]_n = [\mu' \Omega (\mathbf{I} - \Omega)^{-1}]_n$$

- ▶ the cumulative response of aggregate output to a unit shock to n , net of n 's impact
 - ▶ large if n supplies, directly or indirectly, large sectors

The traffic through n is the product of the two

- A walk through $n =$ a walk into n , followed by a walk out of n
- So $b_n \times s_n$ adds up every walk through n , once per passage, weighted by the shock at its start and the size of its end
- Divided by the total effect (CIR, the same sum over *all* walks): the expected number of passages through n

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- A sector that receives much of the shock but supplies no one ($s_n = 0$) has zero traffic: lengthening its delay postpones nothing

The traffic does not depend on the delays

- Neither b_n nor s_n involves the delays: the network sets which walks a shock takes and how much each carries; the delays set only the arrival dates
- So the average duration is *linear* in the delays, with the traffic as coefficients:

$$\mathcal{T} = \sum_n \tau_n v_n, \quad \frac{\partial \mathcal{T}}{\partial \tau_n} = v_n$$

- A duration is long when walks pass through long delays (guided missiles), or pass through the same sectors many times (cycles: the effect comes back in echoes, next section)

⇒ One more month of delay in sector n adds v_n months to the average duration

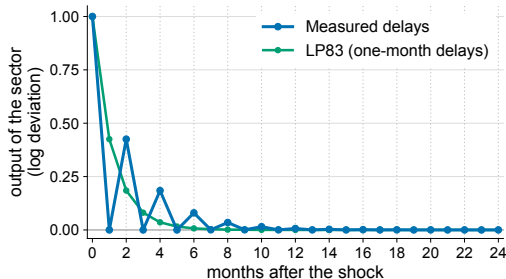
Appendix: bottlenecks (whose delays matter the most), welfare cost of the delays

Bottlenecks

Welfare

Echoes and Endogenous Fluctuations

Nonferrous metal smelting echoes every two months



- Period: 2 months, the sector's delay
- Each wave is about 43% of the previous one = the sector's input share of its own output
- LP83: smooth decline, same total

Output after a shock to the sector's own productivity

- *Self-loop*: a sector that uses its own output, the simplest cycle
- Recall: a cycle's *duration* = sum of the delays along it; its *weight* = product of the input shares along it

⇒ Period = duration of the self-loop; ratio of successive waves = its weight

Does the whole economy behave like this sector? The eigenvalues

- Write the $\text{VAR}(\tau_{\max})$ as a $\text{VAR}(1)$ by stacking the lags:

$$\underbrace{\begin{pmatrix} \hat{\mathbf{y}}_t \\ \hat{\mathbf{y}}_{t-1} \\ \vdots \\ \hat{\mathbf{y}}_{t-\tau_{\max}+1} \end{pmatrix}}_{\equiv \mathbf{Y}_t} = \underbrace{\begin{pmatrix} \mathbf{\Omega}_1 & \mathbf{\Omega}_2 & \cdots & \mathbf{\Omega}_{\tau_{\max}} \\ \mathbf{I} & \mathbf{0} & \cdots & \mathbf{0} \\ & \ddots & \ddots & \vdots \\ \mathbf{0} & \cdots & \mathbf{I} & \mathbf{0} \end{pmatrix}}_{\equiv \mathbf{0}} \underbrace{\begin{pmatrix} \hat{\mathbf{y}}_{t-1} \\ \hat{\mathbf{y}}_{t-2} \\ \vdots \\ \hat{\mathbf{y}}_{t-\tau_{\max}} \end{pmatrix}}_{\equiv \mathbf{Y}_{t-1}} + \underbrace{\begin{pmatrix} \hat{\mathbf{A}}_t \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{pmatrix}}_{\equiv \mathbf{e}_t}$$

- Response t months after a shock $\mathbf{e}_0$: $\mathbf{Y}_t = \mathbf{0}^t \mathbf{e}_0$

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- Response t months after a shock $\mathbf{e}_0$: $\mathbf{Y}_t = \mathbf{O}^t \mathbf{e}_0$
- Write the shock in the basis of the eigenvectors $\mathbf{v}_j$ of $\mathbf{O}$, $\mathbf{e}_0 = \sum_j a_j \mathbf{v}_j$. Since $\mathbf{O} \mathbf{v}_j = \xi_j \mathbf{v}_j$, each application of $\mathbf{O}$ multiplies the j -th piece by its eigenvalue:

$$\mathbf{Y}_t = \mathbf{O}^t \mathbf{e}_0 = \sum_j \xi_j^t a_j \mathbf{v}_j$$

- Each eigenvalue ξ_j contributes one term ξ_j^t to the response; the shock sets only its size a_j (a repeated eigenvalue adds a polynomial factor in t , which does not change the period)

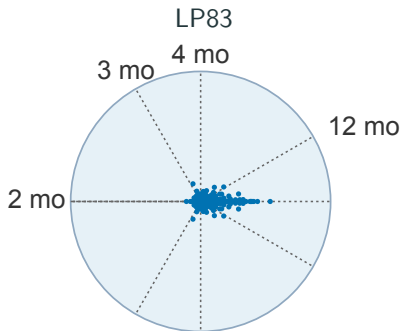
What one eigenvalue does to the response

- Polar form $\xi = |\xi|e^{i\theta}$, so $\xi^t = |\xi|^t e^{i\theta t}$
- **Modulus** $|\xi| < 1$: the term shrinks by $|\xi|$ every month.
- **Argument** θ : whether the term oscillates, and at which period
 - ▶ $\theta = 0$ (positive real): smooth decay, no oscillation
 - ▶ $\theta = \pi$ (negative real): the sign flips every month, a period of 2 months
 - ▶ complex pair: a real oscillation with a period of $2\pi/\theta$ months
- The economy *oscillates* at the periods of its eigenvalues that are **not positive real**
- Static network: all eigenvalues zero, no dynamics. LP83: $\mathbf{O} = \mathbf{\Omega}$

The eigenvalues of O

Each dot: one eigenvalue. Distance to the center = modulus; rays = periods $2\pi/\theta$.

Histogram

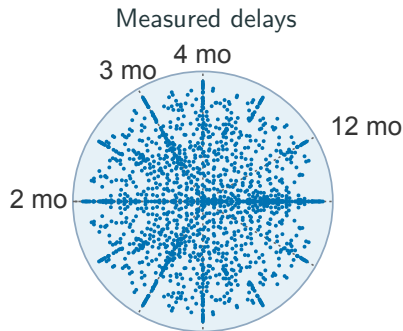
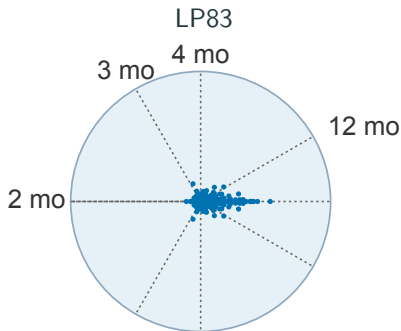


- LP83: small, near the positive real axis: shocks die out quickly and smoothly

The eigenvalues of $\mathbf{O}$

Each dot: one eigenvalue. Distance to the center = modulus; rays = periods $2\pi/\theta$.

Histogram



- LP83: small, near the positive real axis: shocks die out quickly and smoothly
- Measured delays: larger (up to 0.92), along rays at periods of 2, 3, 4 and 12 months

⇒ Same input-output matrix in both. Where do the oscillations come from?

Where the eigenvalues come from: one cycle

- Take a sector n on a single cycle: its output comes back to it along one route only, with **weight** w and **duration** τ
- Shock to n at date 0. After impact, n moves again only when an earlier movement comes back around the cycle: τ months later, multiplied by w

$$\hat{y}_{n,t} = w \hat{y}_{n,t-\tau} \quad (t > 0)$$

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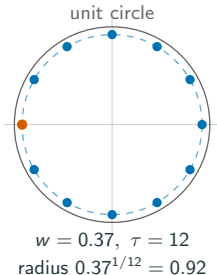
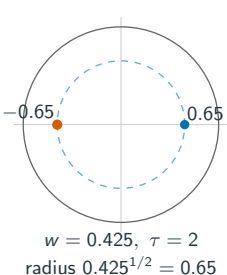
- Recall: the response is a sum of terms ξ^t , one per eigenvalue
- The equation is linear, so each term must satisfy it on its own (distinct geometric sequences cannot offset each other at every date). Plug in $\hat{y}_{n,t} = \xi^t$:

$$\xi^t = w \xi^{t-\tau} \quad \Longleftrightarrow \quad \boxed{\xi^\tau = w}$$

- So the cycle's weight and duration determine its eigenvalues

The τ roots of $\xi^\tau = w$

- $\xi^\tau = w$ has τ solutions, evenly spaced on a circle: $\xi_k = w^{1/\tau} e^{2\pi i k / \tau}$, $k = 0, \dots, \tau - 1$
- Same **modulus** $w^{1/\tau}$: every term shrinks by w every τ months, the ratio between successive waves
- **Arguments** $2\pi k / \tau$: periods τ/k
 - ▶ $k = 0$: positive real, no oscillation
 - ▶ $k = 1$: the period is τ , the duration of the cycle



- Nonferrous metal: $\xi^2 = 0.425$, $\xi = \pm 0.65$; the root -0.65 is the 2-month echo of nonferrous metal. Propulsion units: twelve roots of modulus 0.92, the slowest to die out
- In LP83 the same self-loops have $\tau = 1$: one root, $\xi = w$, smooth decay

More on eigenvalues

Proposition (eigenvalues and cycles)

- (i) All eigenvalues of $\mathbf{O}$ are zero iff the network is vertical (has no cycle).*
- (ii) If a cycle with weight w and duration τ is isolated (its sectors lie on no other cycle), the τ roots of $\xi^\tau = w$ are eigenvalues of $\mathbf{O}$.*
- (iii) General case, any sector, in the paper.*

General case

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General case

Why LP83 barely oscillates

- In LP83 $\mathbf{O} = \mathbf{\Omega}$: a self-loop gives one eigenvalue, ω_{nn} , positive real
- Its oscillations can only come from the eigenvalues of $\mathbf{\Omega}$ that are not positive real, and those are small (Liu and Tsyvinski, 2024)
- With a delay $\tau_n \geq 2$ the same self-loop gives τ_n roots, and all but one oscillate

⇒ Same input-output matrix: the delays turn its cycles into oscillations

The self-loops are visible in the U.S. eigenvalues

Why they should be

Almost no cycle is isolated: 345 of 371 sectors are connected by cycles

but a large self-loop is by far the heaviest walk back to its sector

so its roots, $\xi^{\tau_n} = \omega_{nn}$, should be close to eigenvalues

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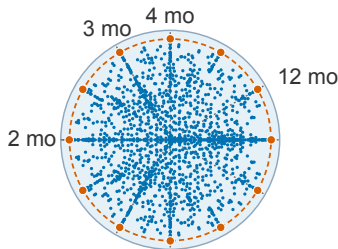
Almost no cycle is isolated: 345 of 371 sectors are connected by cycles

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so its roots, $\xi^{\tau_n} = \omega_{nn}$, should be close to eigenvalues

What we see

- Top ten self-loops: every one of their roots is within 2.2% of an eigenvalue
- Rays: self-loops with the same delay put their roots on the same ray; heavier, farther out
- Outer ring at 0.92 (orange): the twelve roots of the propulsion-units self-loop, an annual echo
 - ▶ weight 0.37, delay 12 months: $0.37^{1/12} = 0.92$



Eigenvalues, measured delays

⇒ The eigenvalues that die out most slowly are the ones the self-loops predict

Why second moments, when we have the eigenvalues

- Eigenvalues: which **periods** exist, and which cycles create them
 - ▶ the modulus gives a term's **decay**, not its **amplitude** in a given sector
 - ▶ not something we can **measure**

Why second moments, when we have the eigenvalues

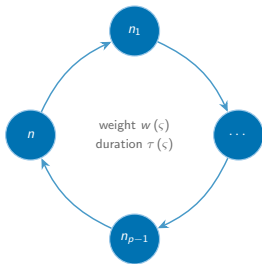
- Eigenvalues: which **periods** exist, and which cycles create them
 - ▶ the modulus gives a term's **decay**, not its **amplitude** in a given sector
 - ▶ not something we can **measure**
- Autocovariances: the same cycles, **sector by sector**, with sizes, and measurable
 - ▶ $\gamma_k(n) = E[\hat{y}_{nt}\hat{y}_{n,t-k}]$
 - ▶ we never see one impulse response: new shocks arrive every month
- Spectrum: the variance at the period $2\pi/\omega$ months

$$f_n(\omega) = \frac{1}{2\pi} \sum_k \gamma_k(n) e^{-i\omega k}$$

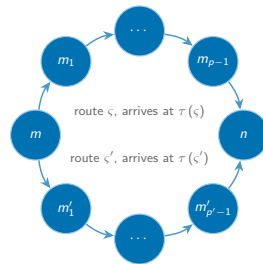
- ▶ a peak every τ months in $\gamma_k \Rightarrow$ spectral peaks at $2\pi/\tau$ and its multiples (ω : a frequency, not an input share)

Where serial correlation comes from

- i.i.d. shocks: $\hat{y}_{nt}$ and $\hat{y}_{n,t-k}$ covary only if the **same shock** reaches n at both dates
- So: **two walks** from the shocked sector to n , arriving k months apart



(a) **Directed cycle**: same route, extra laps
Correlation at **every multiple** of the duration
(also for sectors downstream of the cycle)



(b) **Undirected cycle**: two routes
Correlation at **one lag**, $|\tau(\zeta) - \tau(\zeta')|$

$\Rightarrow$ Only directed cycles make serial correlation recur

A cycle leaves a comb in the autocovariances

- A cycle of weight w and duration τ through sector n
- After k laps, n 's output contains w^k times its own value $k\tau$ months earlier
- The other pairs of walks only add to this (nonnegative terms)

Proposition (contribution of a cycle, i.i.d. shocks)

For a cycle of weight w and duration τ through sector n ,

$$\gamma_{k\tau}(n) \geq w^{|k|} \sigma_n^2 \quad \text{for every } k, \quad \sigma_n^2 = \text{var}(\hat{y}_{nt})$$

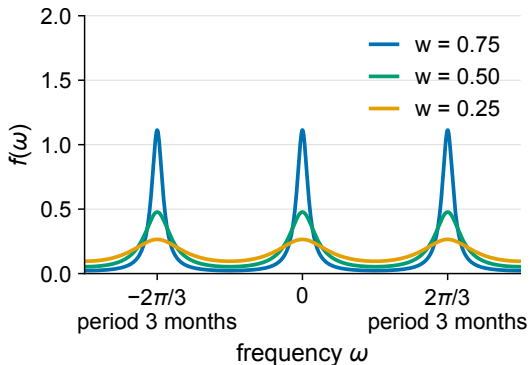
- A **comb**: a peak every τ months, the k -th at least $w^k \sigma_n^2$
- A sector on several cycles: their combs **add up**, still a lower bound

The same comb in the spectrum

Proposition (contribution of a cycle, continued)

If the cycle were the only source of serial correlation of sector n , its spectrum would be

$$\frac{\sigma_n^2}{2\pi} \frac{1 - w^2}{1 + w^2 - 2w \cos(\omega\tau)}$$

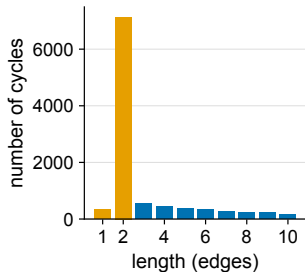


- Peaks where $\cos(\omega\tau) = 1$: frequency $2\pi/\tau$ (period τ) and its multiples.
Figure: $\tau = 3$
- Heavier cycle: **sharper** peaks
- Several cycles: the peaks of each **add up**

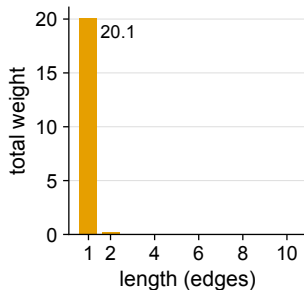
Harmonics

Poisson

The heavy cycles of the U.S. network are self-loops



(a) Cycles by length



(b) Total weight w , by length

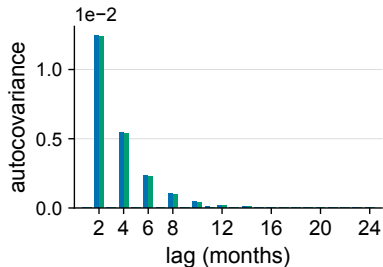
- Many long cycles
- **Weight: almost all on self-loops**
- Largest: nonferrous metal 0.43, beef cattle 0.41, distilleries 0.39
- Two-sector cycles: at most 0.025
- Self-loop above 0.1, a strong echo: **one sector in six** (13% of output). Median 0.02

■ lengths 1 and 2, all listed; ■ longer cycles, found by 250 random crawlers

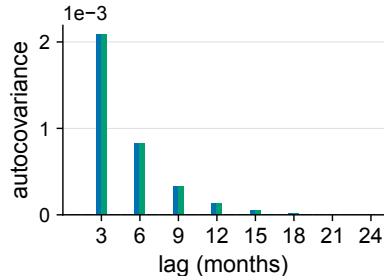
The cycles account for the serial correlation of the model

- The proposition is a **lower bound**. How tight?
- Per sector: sum $w^k \sigma_n^2$ over its **directed** cycles; compare with the model at lags 1–60
- Pooled $R^2 = 0.97$, median sector 0.85. Self-loops alone: the same

Lags 1 to 24 months: ■ autocovariances of the model; ■ sum of the directed-cycle contributions



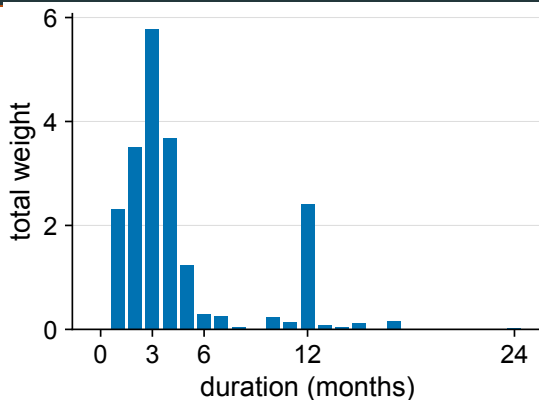
(a) Nonferrous metal: self-loop 0.43, 2 months



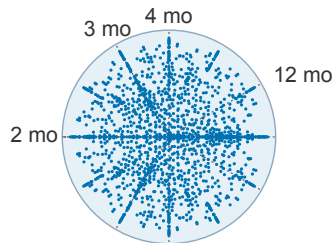
(b) Distilleries: self-loop 0.39, 3 months

⇒ The bound is nearly the whole story, and the self-loops are key

The durations of the heavy cycles are the rays of the eigenvalues



Total cycle weight, by duration



Eigenvalues, measured delays

- Cycle weight at 2, 3, 4 and 12 months; eigenvalue rays at 2, 3, 4 and 12 months
- Caveat: 84% of the 4-month weight is sectors with no backlog data, set to 4
- LP83: every self-loop lasts one month, so no rays

Sectoral Comovements and Aggregation

From sectors to aggregate output

- Aggregate output: $\hat{y}_t = \sum_n \mu_n \hat{y}_{nt}$, μ_n = value-added share (as in industrial production)
- Its serial correlation has **two sources**
 - ▶ each sector's own: the **cycles** of the previous section
 - ▶ **comovement** between sectors
- Comovement of n and m at lag l , with i.i.d. shocks:

$$[\Gamma_l]_{nm} = \sum_j \sigma^2(\hat{A}_j) \sum_{\substack{\varsigma: j \rightarrow n, \varsigma': j \rightarrow m \\ \tau(\varsigma) - \tau(\varsigma') = l}} w(\varsigma) w(\varsigma')$$

- $\Gamma_l = E[\hat{\mathbf{y}}_t \hat{\mathbf{y}}'_{t-l}]$: each pair of walks from a common source j , arriving l months apart, adds the product of its weights times the variance of the shock to j

⇒ Do the echoes survive aggregation?

Two sources of aggregate fluctuations

Proposition

The spectrum of aggregate output is

$$f_y(\omega) = \underbrace{\sum_{n=1}^N \mu_n^2 f_n(\omega)}_{\text{sum of sectoral spectra}} + \underbrace{\frac{1}{2\pi} \sum_{n \neq m} \sum_{k=-\infty}^{\infty} \mu_n \mu_m [\Gamma_k]_{nm} e^{-i\omega k}}_{\text{sectoral comovement term}}$$

with $\Gamma_k = E[\hat{\mathbf{y}}_t \hat{\mathbf{y}}'_{t-k}]$ the autocovariances of sectoral output.

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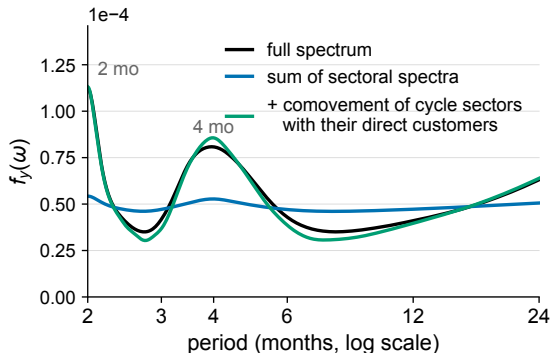
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with $\Gamma_k = E[\hat{\mathbf{y}}_t \hat{\mathbf{y}}'_{t-k}]$ the autocovariances of sectoral output.

- **Sectoral spectra**: each has peaks at its cycles' durations; 371 small weights **mute** them, but cannot cancel them
- **Comovement**: pairs of walks from a common shock
 - ▶ a cycle sector and its customers echo **together**, at the cycle's duration
 - ▶ so this term peaks at the same periods and brings them back

The echoes survive aggregation



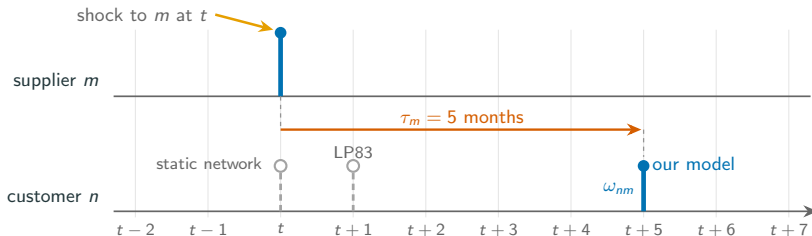
- 371 sectors with small weights could average the echoes away
- They do not: peaks at 4 and 2 months many self-loops
- Each sector: its directed cycles explain its own serial correlation
- The cycles' echoes reach the aggregate through their customers, who echo with them (green)

The 4-month peak partly reflects the default delay of sectors without backlog data.

What the delays do to the business cycle

Empirical Evidence

A supplier's shock reaches its customers exactly one delay later



Law of motion

$$\hat{y}_{nt} = \hat{A}_{nt} + \sum_m \omega_{nm} \hat{y}_{m,t-\tau_m}$$

In growth rates

$$\Delta \hat{y}_{nt} = \Delta \hat{A}_{nt} + \underbrace{\sum_m \omega_{nm} \Delta \hat{y}_{m,t-\tau_m}}_{X_{nt}^{(\tau)}, \text{ the delayed composite}}$$

- No free parameter: the shares, the delays and the outputs are all measured

Bringing the model to the data

- We need high-frequency (monthly) output (quantity) by industry
 - ▶ Fed's industrial prod. indexes (G.17), 104 ind., 1972–2026, matched to BEA sectors
- Input shares ω from the BEA tables, delays τ from the backlog data.
- Two things the G.17 does not give us
 - ▶ **all suppliers**: it covers manuf., mining and utilities, about 2/3 of inputs for typical ind.
 - ▶ **a clean date for output**: some indexes record output when it is built, others when it is sold

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 - ▶ **all suppliers**: it covers manuf., mining and utilities, about 2/3 of inputs for typical ind.
 - ▶ **a clean date for output**: some indexes record output when it is built, others when it is sold
- Our answer: simulate each economy and measure it **exactly like the data**
 - ▶ 100 panels per economy: same missing suppliers, same recording dates, same regression
 - ▶ An economy's prediction: the distribution of its 100 estimates

How the G.17 dates output

- The model's $\hat{y}_{nt}$ is output **produced** in month t
- **Built-dated** indexes measure exactly that: output recorded while it is produced (hours, electricity). About 45% of value added
- **Sale-dated** indexes: output measured at shipment or completion. About 55%

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- **Sale-dated** indexes: output measured at shipment or completion. About 55%
- A sale-dated index has two possible timings
 - ▶ as the work progresses (the Census instruction to respondents)
 - ▶ nothing is recorded until delivery, a full delay later
- Which one holds is **unknown**, industry by industry

Details

The test: regress output growth on suppliers' growth, lagged by their delays

- **Delayed composite** $X_{nt}^{(\tau)}$: the suppliers' growth, lagged by their delays
- **Contemporaneous composite** $X_{nt}^{(0)}$: the same suppliers, no lag

$$\Delta \hat{y}_{nt} = \beta_{\tau} X_{nt}^{(\tau)} + \beta_0 X_{nt}^{(0)} + \alpha_n + \delta_t + s_{n,mo(t)} + \varepsilon_{nt}$$

Fixed effects: industry, month, industry $\times$ calendar month

	Our model (measured delays)	Static network	LP83
Predicted β_{τ}	1	0	≈ 0
because	law of motion in growth rates	all comovement is today	all arrives at lag one

- With random-walk productivity these predictions are exact; the simulations use the estimated persistence

Proposition

Customers move with their suppliers at the measured delays

- **Dating-adjusted** design: shift sale-dated customers back, drop sale-dated suppliers

Data		Our model, sale-dated recorded		Static	LP83
$\hat{\beta}_\tau$	(s.e.)	as the work progresses	at delivery		
0.91	(0.49)	0.62 ± 0.09	0.65 ± 0.10	−0.25 ± 0.08	−0.12 ± 0.09

s.e.: clustered by industry. ±: spread over 100 simulated panels. Static and LP83 recorded at delivery (a full measured delay late in the static network, one month in LP83).

Customers move with their suppliers at the measured delays

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s.e.: clustered by industry. $\pm$: spread over 100 simulated panels. Static and LP83 recorded at delivery (a full measured delay late in the static network, one month in LP83).

- Our model: same prediction under either timing; the data are within one s.e.
- Peak at **exactly the measured delay**; beats all 500 random reassignments

⇒ The data are above every static and LP83 draw

Designs

Census-only delays

Peak

Permutations

Two predictions from network cycles

- **Self-loop echo**: n 's growth vs. its own growth τ_n months earlier; correlation of at least ω_{nn}
- **Trading-pair arm**: n and m sell to each other; n 's growth vs. m 's growth τ_m months earlier
- The echo is not automatic: a BEA “self-loop” can be two industries inside one sector buying from each other, with **no echo**. Finding it is informative

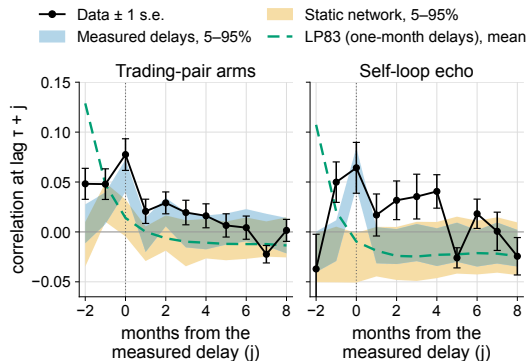
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How we measure

- Correlation of n 's growth with the partner's growth at lag $\tau + j$, for $j = -2, \dots, 8$; averaged over the pairs (or sectors)
- Prediction: a **spike at $j = 0$** . Static network: none. LP83: at lag one, that is at $j = 1 - \tau$, two or more months **before** the delay
- **Excess correlation**: value at $j = 0$ minus the average at $j = \pm 2, \pm 3$

The network's cycles show in the data at the measured delays



	Excess correlation	
	Trading-pair arms	Self-loop echo
Data	+0.045	0.053
(s.e.)	(0.013)	(0.022)
Our model	0.047	0.078
Static	+0.02	≈ 0
LP83	-0.03	-0.03

- Data: a **spike** at $j = 0$ in both panels, inside our model's band
- Static: flat. LP83: elevated at $j \leq -2$ (its lag one), **negative** at the delay

Bands: 5th–95th percentiles of 100 simulated panels. Echo: 29 industries with delays ≥ 3 months, indexes at least 80% built-dated. Arms: 53 pairs with input shares ≥ 0.002 both ways and both delays ≥ 3 months.

Aggregate industrial production follows its suppliers' delays

- Same regression on aggregate industrial production; prediction $\beta_\tau = 1$
- Series used as recorded; **both timings** simulated, as in the panel

	Our model, sale-dated output recorded				
	Data	as the work progresses	at delivery	Static	LP83
$\widehat{\beta}_\tau$	1.18	1.12 ± 0.12	0.12 ± 0.10	0.31 ± 0.13	0.16 ± 0.14

$\pm$: spread over 100 simulated histories. Static at delivery, LP83 as the work progresses (most favorable to each).

- Static and LP83: below 0.35 under either reading

Both readings in full

All tests

Conclusion

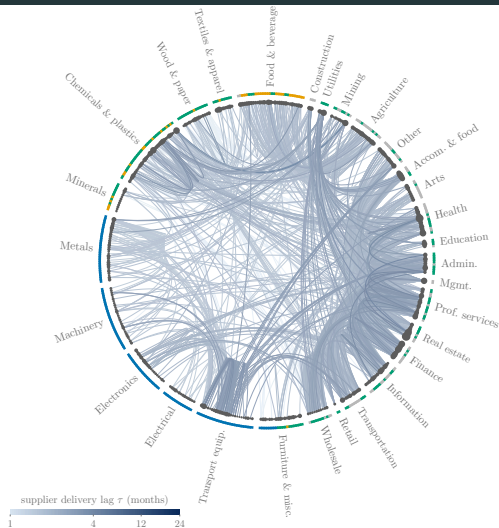
- Delays change **when** a shock arrives, not how much arrives: longer, uneven **persistence** across sectors
- The economy **fluctuates** at the durations of the network's cycles: **echoes**, mostly from self-loops, visible in aggregate output
- **Evidence**: in monthly industrial production, shocks reach customers at the measured delays, with no free parameter. The data favor the model over the static network and LP83

More in the paper

- The welfare cost of delays and the **bottleneck** sectors
- Measured delays **damp** aggregate fluctuations at business-cycle periods
- Comovement across sectors through pairs of walks

Backup

The network and its delays



- Each point is a sector, sized by its Domar weight (its sales over GDP)
- Chords: the 1,500 largest input flows, darker when the supplier's delay is longer
- Outer ring: the source of each sector's delay, colored as in the histogram
- Long delays: durable manufacturing, and project services (engineering, research, computer-systems design)
- Both ship their output throughout the economy

Prices: two markets, one at each end of production

- Prices are the planner's shadow values, in units of the household's utility
- A firm buys its inputs on the **spot** market, for delivery today:

$$p_{nt} = \frac{\partial V}{\partial X_{n0,t}} = \frac{\zeta_n}{X_{n0,t}} \quad \text{value of one more unit of } n \text{ delivered today}$$

- It sells its output **forward**, for delivery τ_n periods later:

$$q_{nt} = \beta E_t \left[\frac{\partial V}{\partial X_{n,\tau_n-1,t+1}} \right] = \beta^{\tau_n} \frac{\zeta_n}{y_{nt}} \quad \text{value today of that future delivery}$$

- ▶ in steady state $q_n = \beta^{\tau_n} p_n$: the spot price, discounted over the wait
- Every nominal magnitude is constant: spot value of the delivered stock $p_{nt} X_{n0,t} = \zeta_n$; forward revenue $q_{nt} y_{nt} = \beta^{\tau_n} \zeta_n$; spending on input $m =$ the share ω_{nm} of revenue; consumption expenditure $\sum_n p_{nt} c_{nt} = \sum_n \gamma_n = 1$

$\Rightarrow$ Forward revenue $\beta^{\tau_n} \zeta_n$ is the marginal social value of a shock to n

Value added, nominal GDP and Domar weights

- Value is created at **production**, when the inputs are used: the national-accounts convention (goods in process count as output when produced)
- Value added = forward revenue – inputs bought at spot prices. Inputs absorb the share $1 - \alpha_n$ of revenue, so

$$VA_n = q_{nt}y_{nt} - \sum_m p_{mt}x_{nm,t} = \alpha_n \beta^{\tau_n} \zeta_n \quad = \text{the wage bill of } n: \text{ labor is the only primary factor}$$

- Nominal GDP = sum of value added, a constant: $GDP^{\text{nom}} = \sum_n \alpha_n \beta^{\tau_n} \zeta_n$
- **Domar weight** = forward sales over nominal GDP. Delays enter only through β^{τ_n} ; without discounting it is the static network's Domar weight:

$$\lambda_n = \frac{q_{nt}y_{nt}}{GDP^{\text{nom}}} = \frac{\beta^{\tau_n} \zeta_n}{\sum_k \alpha_k \beta^{\tau_k} \zeta_k} \xrightarrow{\beta \rightarrow 1} \lambda_n^0 = [(\mathbf{I} - \mathbf{\Omega}')^{-1} \boldsymbol{\gamma}]_n$$

Real GDP and Hulten's theorem

Definition. Real GDP is the chained (Divisia) volume index of value added: double-deflated sectoral real value added, aggregated with value-added shares (constant here, so no chaining is needed):

$$d \log Q_{nt}^{\text{VA}} = \frac{q_{nt} y_{nt}}{\text{VA}_{nt}} d \log y_{nt} - \sum_m \frac{p_{mt} x_{nm,t}}{\text{VA}_{nt}} d \log x_{nm,t}, \quad d \log Y_t = \sum_n \frac{\text{VA}_{nt}}{\text{GDP}^{\text{nom}}} d \log Q_{nt}^{\text{VA}}$$

Proposition

Real GDP growth is the Domar-weighted sum of productivity growth:

$$d \log Y_t = \sum_n \lambda_n d \log A_{nt}$$

- Hulten's theorem survives heterogeneous time-to-build. With i.i.d. shocks it also measures welfare: $dV = \sum_n q_{nt} y_{nt} d \log A_{nt} = \text{GDP}^{\text{nom}} \times d \log Y_t$ (a 1% shock to n adds 1% of its output, worth its forward price)
- Real GDP is **blind to the delays**: value added is counted at production, so the extra output counts today, as goods in process; with i.i.d. shocks it is white noise

Domar Weights



Top 20 sectors by ζ_n , the value of a delivered good (spot sales; the axis label). The model's Domar weight discounts it for the wait: $\lambda_n = \beta^{\tau_n} \zeta_n / \text{GDP}^{\text{nom}}$.

Prices and GDP

◀ Back

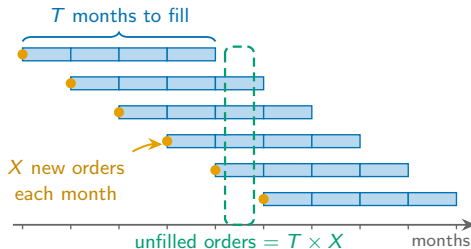
Three aggregates see a shock at different times

With $\mathbf{M}(z) = (\mathbf{I} - \mathbf{\Omega} \mathbf{diag}(z^\tau))^{-1}$, the sum of $w(\varsigma)z^{\tau(\varsigma)}$ over walks, output is $\hat{\mathbf{y}}_t = \mathbf{M}(L)\hat{\mathbf{A}}_t$.

- **Real GDP** (chained index of value added): $d \log Y_t = \sum_n \lambda_n d \log A_{nt}$. With i.i.d. shocks it is white noise, because value added is counted at production. It is blind to the delays
- **Consumption**: $\hat{C}_t = \gamma' \mathbf{M}^D(L) \hat{\mathbf{A}}_t$, $\mathbf{M}^D(z) = \mathbf{diag}(z^\tau) \mathbf{M}(z)$: one more delivery leg, so nothing on impact; its discounted sum is the welfare value of the shock
- **Aggregate output**: $\hat{y}_t = \mu' \mathbf{M}(L) \hat{\mathbf{A}}_t$, $\mu_n = \text{VA}_n / \text{GDP}^{\text{nom}} = \alpha_n \lambda_n$: value-added-weighted gross output, what industrial production measures. It carries the full propagation dynamics

⇒ In the static network all three move on impact; delays separate their timing

The backlog ratio measures the delay



- Backlog ratio = unfilled orders / monthly shipments = months needed to clear the order book. In the model's steady state it equals the delay exactly
- Measured for 296 of the 371 sectors, 53.5 percent of value added; the rest receive the average of the measured manufacturing sectors, 4 months

Further related literature

- **Growth accounting with intermediate inputs:** Hulten (1978)
- **Distortions in production networks:** Jones (2011), Liu (2019), Baqaee and Farhi (2019)
- **Delays in supply chains:** Meier (2020)

Average duration: differentiate the walk sum at $z = 1$

- Response of aggregate output $\sum_n \mu_n \hat{y}_{nt}$ to a shock $\hat{\mathbf{A}}$ at date 0, as a power series:

$$W(z) = \sum_{t \geq 0} z^t \sum_n \mu_n \hat{y}_{nt} = \boldsymbol{\mu}' \mathbf{M}(z) \hat{\mathbf{A}}, \quad \text{CIR} = W(1), \quad \mathcal{T} = \frac{W'(1)}{W(1)}$$

- $\mathbf{M}'(1) = \mathbf{M}(1) \boldsymbol{\Omega} \text{diag}(\boldsymbol{\tau}) \mathbf{M}(1)$ with $\mathbf{M}(1) = (\mathbf{I} - \boldsymbol{\Omega})^{-1}$. Writing $\text{diag}(\boldsymbol{\tau}) = \sum_n \tau_n \boldsymbol{\delta}_n \boldsymbol{\delta}_n'$:

$$\mathcal{T} = \sum_n \tau_n \frac{s_n b_n}{\text{CIR}}, \quad b_n = \boldsymbol{\delta}_n' (\mathbf{I} - \boldsymbol{\Omega})^{-1} \hat{\mathbf{A}}, \quad s_n = \boldsymbol{\mu}' \boldsymbol{\Omega} (\mathbf{I} - \boldsymbol{\Omega})^{-1} \boldsymbol{\delta}_n$$

- Walk reading: a walk arrives τ_n later for each passage through n (its last sector excluded). $b_n s_n$ adds every walk once per passage through n , so $v_n = s_n b_n / \text{CIR}$ is the expected number of passages: the traffic
- The traffic does not depend on the delays, so the duration is linear in them

Bottlenecks: the sectors whose delays matter the most

- A supply disruption or a logistics failure adds months to one sector's delay. We ask which sectors slow the economy the most when that happens
- $\mathcal{T}$ is linear in the delays, with the traffic as coefficients:

Corollary (bottlenecks)

One more month in sector n 's delay raises the average duration by the traffic through n :

$$\partial \mathcal{T} / \partial \tau_n = v_n = b_n s_n / \text{CIR}$$

- A bottleneck needs both factors: central as a buyer *and* as a supplier
- Largest traffic: other real estate, insurance agencies, management of companies, basic organic chemicals
- Size is not enough: owner-occupied housing and hospitals, the two largest sectors by Domar weight, rank 356th and 180th. They end supply chains: almost nothing passes through them to another sector
- The traffic does not depend on the delays, so the ranking does not use the delay data

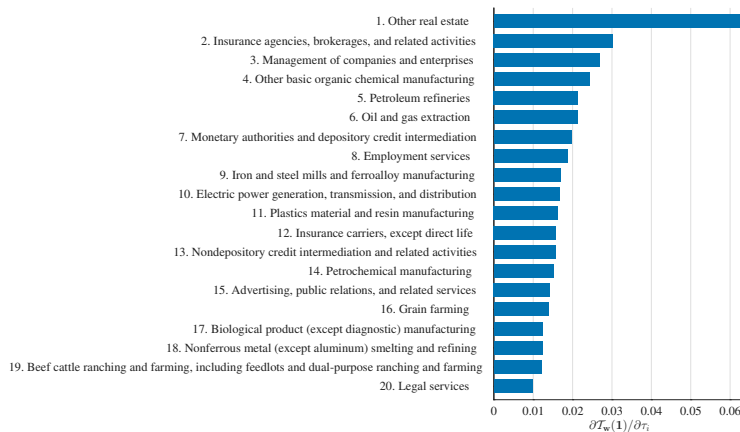
Propagation bottlenecks are input hubs; welfare bottlenecks are large sectors

v_n, v_n^W : rise in the average duration $\mathcal{T}$ and the welfare duration $\mathcal{T}^W$ (unit shock to every sector) per extra month of delay in n ; λ_n : Domar weight.

Propagation bottlenecks				Welfare bottlenecks			
Sector	v_n	λ_n	rank as welfare bottleneck	Sector	v_n^W	λ_n	rank as propagation bottleneck
Other real estate	0.064	0.057	3	Owner-occupied housing	0.079	0.120	356
Insurance agencies	0.030	0.025	16	Hospitals	0.078	0.082	180
Management of companies	0.027	0.033	10	Other real estate	0.066	0.057	1
Basic organic chemicals	0.024	0.013	22	Petroleum refineries	0.051	0.039	5
Petroleum refineries	0.021	0.039	4	Insurance carriers	0.039	0.040	12

- Propagation ($v_n \propto s_n b_n$): central buyer *and* supplier; size is not enough (housing, hospitals)
- Welfare ($v_n^W \propto \lambda_n^0 b_n$, the household counts as a customer): mostly size; 7 of the 8 largest sectors by Domar weight are in the top 8
- The traffic does not depend on the delays, so the ranking does not use the delay data

Bottlenecks



Top 20 propagation bottlenecks: rise in the average duration of aggregate output per extra month of delay in the sector, for a unit shock to every sector.

Table

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The welfare cost of delays: the welfare duration

- The household discounts the future: gains that arrive later are worth less
- The household consumes delivered goods: $\hat{c}_{nt} = \hat{y}_{n,t-\tau_n}$
- So the effect of a shock on consumption is its effect on output, delayed once more by the producing sector's own delay
- **Welfare duration** $\mathcal{T}^W$: average duration of the effect on consumption, with the consumption shares γ_n as weights
- Unit shock to every sector: **9.0 months**, against 2.0 in LP83
 - ▶ longer than the durations of output (4.0 and 0.9): one more delay

The welfare cost of delays: the value of a shock

Proposition (welfare value of a shock)

$$\Delta V = \underbrace{\lambda^{0'} \hat{\mathbf{A}}}_{\text{value in the static network}} \times \underbrace{E[\beta^D]}_{\text{discount over the wait}} = \lambda^{0'} \hat{\mathbf{A}} (1 - r\mathcal{T}^W + O(r^2))$$

- λ^0 : Domar weights of the static network (Hulten's theorem: the first factor is the value of the shock without delays)
- D : the random month in which the effect reaches the household, $E[D] = \mathcal{T}^W$; r : monthly discount rate
- The delays cost the share $1 - E[\beta^D] \approx r\mathcal{T}^W$ of the static value: 2.9% at 4% a year, against two thirds of a percent in LP83
- Shock to guided missiles: welfare duration 56.5 months, 17% of the value lost (exact share; the first-order $r\mathcal{T}^W$ gives 18.5%)
- Welfare bottlenecks: the household counts as a customer, so $\partial\mathcal{T}^W/\partial\tau_n \propto \lambda_n^0 b_n$: mostly size; housing and hospitals lead

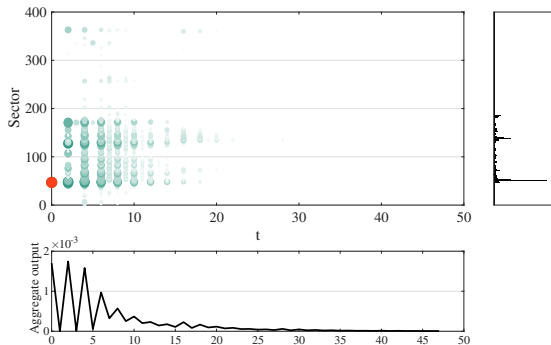
Autocovariances of the VAR(1)

- $\mathbf{Y}_t = \mathbf{O}\mathbf{Y}_{t-1} + \mathbf{e}_t$ with $\mathbf{\Sigma}_e = E[\mathbf{e}_t\mathbf{e}_t']$ and $\mathbf{\Gamma}_k = E[\mathbf{Y}_t\mathbf{Y}_{t-k}']$:

$$\mathbf{\Gamma}_0 = \sum_{k=0}^{\infty} \mathbf{O}^k \mathbf{\Sigma}_e (\mathbf{O}')^k, \quad \mathbf{\Gamma}_k = \mathbf{O}^k \mathbf{\Gamma}_0$$

- The first $N \times N$ block of $\mathbf{\Gamma}_k$ is the sectoral $E[\hat{\mathbf{y}}_t \hat{\mathbf{y}}_{t-k}']$: its (n, m) entries are the comovements, its (n, n) entries the autocovariances $\gamma_k(n)$ of sector n
- This computes every sector's autocovariances and spectrum, but says nothing about where they come from: the cycles do

A shock to nonferrous metal smelting spreads in two-month waves



Shock of 1% to sector #331410 (red dot at $t = 0$). Top: response of each of the 371 sectors by month (dot size and shade = magnitude, log scale); right: cumulated response by sector. Bottom: aggregate output, in percent.

Eigenvalues and cycles: the general statement

$\mathbf{O}$ = companion matrix of the VAR; a response is $\sum_j \xi_j^t c_j$ over its eigenvalues ξ_j . A first-return walk ς of n leaves n and comes back to it for the first time; the *first-return function* is $R_n(z) = \sum_{\varsigma} w(\varsigma) z^{\tau(\varsigma)}$.

Proposition

(i) All eigenvalues of $\mathbf{O}$ are zero iff the network is vertical (has no cycle). (ii) For any sector n , every solution ξ of $R_n(\xi^{-1}) = 1$ is an eigenvalue of $\mathbf{O}$. (iii) An isolated cycle (its sectors lie on no other cycle) of weight w and duration τ contributes the τ roots $\xi_k = w^{1/\tau} e^{2\pi i k / \tau}$, $k = 0, \dots, \tau - 1$.

- Why (ii): after impact, $\hat{y}_{n,t} = \sum_{\varsigma} w(\varsigma) \hat{y}_{n,t-\tau(\varsigma)}$, and ξ^t solves it iff $R_n(\xi^{-1}) = 1$
- (iii) is the special case of (ii) with $R_n(z) = wz^\tau$: the cycle is n 's only first-return walk
- U.S.: 345 of 371 sectors are connected by cycles; six self-loops are isolated

Complex eigenvalues give a real response

- $\mathbf{O}$ is real, so its complex eigenvalues come in conjugate pairs: if $\mathbf{O}\mathbf{v} = \xi\mathbf{v}$, then $\mathbf{O}\bar{\mathbf{v}} = \bar{\xi}\bar{\mathbf{v}}$
- The shock is real, so the two pieces get conjugate sizes a and $\bar{a}$
- Their sum is real:

$$a\xi^t\mathbf{v} + \bar{a}\bar{\xi}^t\bar{\mathbf{v}} = 2\operatorname{Re}(a\xi^t\mathbf{v})$$

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$$a \xi^t \mathbf{v} + \bar{a} \bar{\xi}^t \bar{\mathbf{v}} = 2 \operatorname{Re} (a \xi^t \mathbf{v})$$

- For sector n , with $\xi = |\xi|e^{i\theta}$:

$$2 |a v_n| |\xi|^t \cos(\theta t + \varphi_n), \quad \varphi_n = \arg(a v_n)$$

- A damped wave with period $2\pi/\theta$, the same for every sector
- The phase φ_n depends on the sector: sectors oscillate at the same period but can peak in different months

$\Rightarrow$ Each conjugate pair is one real damped wave

A cycle of duration τ has τ eigenvalues

- On the cycle, after the shock: $\hat{y}_{n,t} = w \hat{y}_{n,t-\tau}$
- To know the future, we need the sector's output in each of the last τ months: those goods are still on their way back to it
- So the cycle has τ state variables, and τ eigenvalues

A cycle of duration τ has τ eigenvalues

- On the cycle, after the shock: $\hat{y}_{n,t} = w \hat{y}_{n,t-\tau}$
- To know the future, we need the sector's output in each of the last τ months: those goods are still on their way back to it
- So the cycle has τ state variables, and τ eigenvalues
- Propulsion units: $\tau = 12$, twelve eigenvalues
- LP83: $\tau = 1$, one state variable, one eigenvalue

⇒ The number of roots counts the months the cycle has to remember

Next

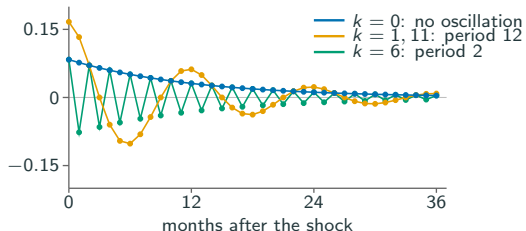
◀ Back

The twelve terms add up to one echo a year

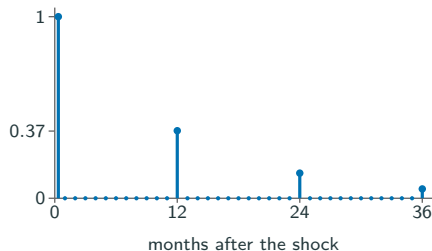
- Shock at date 0: response 1, then w at month 12, w^2 at month 24, zero in between
- One term ξ^t never stays at zero for eleven months; the twelve terms together do:

$$\frac{1}{12} \sum_{k=0}^{11} \xi_k^t = w^{t/12} \cdot \frac{1}{12} \sum_{k=0}^{11} e^{2\pi i k t / 12} = \begin{cases} w^{t/12} & \text{if 12 divides } t \\ 0 & \text{otherwise} \end{cases}$$

Three of the twelve terms



Sum of all twelve terms



Each root is a harmonic of the yearly echo

- Root k has argument $2\pi k/\tau$: period τ/k months

The twelve roots of propulsion units

k	0	1, 11	2, 10	3, 9	4, 8	5, 7	6
Period (months)	none	12	6	4	3	2.4	2

- One positive real root, one negative real root, five conjugate pairs: seven periods
- $k = 1$ is the cycle's own period; the others are its harmonics
- They are the peaks of the cycle's spectrum, at τ/k

⇒ One cycle, one echo a year, seven periods in the spectrum

Long delays put the eigenvalues near the unit circle

- All τ roots have modulus $w^{1/\tau}$
- Per lap of the cycle, the response shrinks by w , whatever the delay
- Per month, it shrinks by $w^{1/\tau}$: the longer the delay, the closer to 1

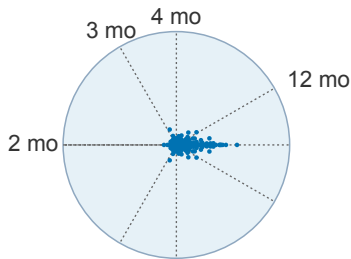
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- All τ roots have modulus $w^{1/\tau}$
- Per lap of the cycle, the response shrinks by w , whatever the delay
- Per month, it shrinks by $w^{1/\tau}$: the longer the delay, the closer to 1
- Propulsion units, $w = 0.37$:
 - ▶ with $\tau = 1$ (LP83) the root would be 0.37: the response loses 63% a month
 - ▶ with $\tau = 12$ the modulus is 0.92: it loses 8% a month
- 0.92 is the largest modulus in the U.S. economy: these echoes die out last

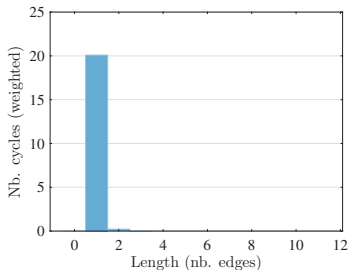
⇒ Delays do not change the loss per lap; they stretch each lap over τ months

From Network Cycles to Business Cycles (LP83)

Eigenvalues, LP83



Cycles by duration in LP83
(weighted)

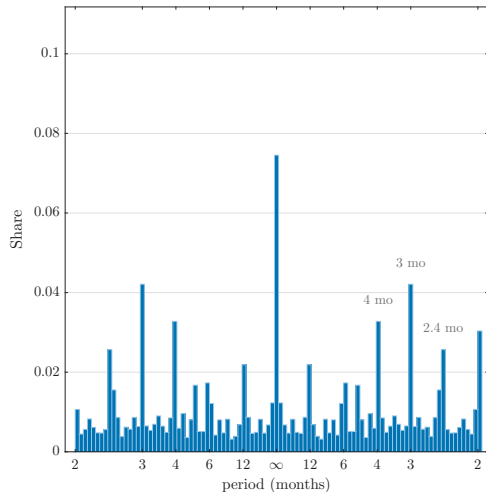


- Every delay is one month: a cycle's duration is its number of links
- Almost all the weight is at duration 1, the self-loops

- A one-month self-loop has one root, $\xi = \omega_{nn}$: real and positive, it does not oscillate
- The companion matrix is Ω itself; its eigenvalues that are not positive real are small (Liu and Tsyvinski)

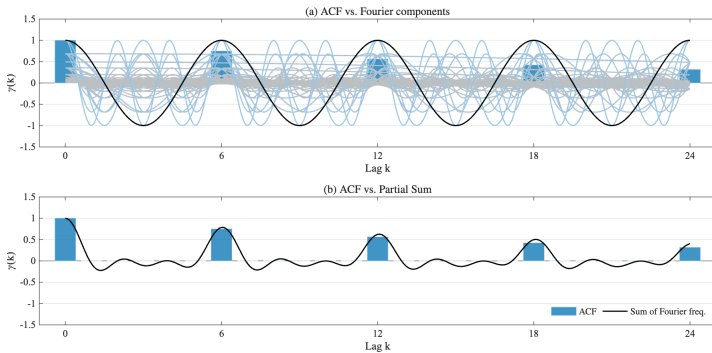
⇒ LP83 puts its cycles at a duration of one month, so it barely oscillates

The periods of the eigenvalues



- Eigenvalues of the companion matrix, measured delays, by argument expressed as a period (months), weighted by modulus
- Center (∞): positive real eigenvalues, which do not oscillate
- Mass at 2, 3, 4 and 12 months, the durations of the heavy self-loops
- A cycle of duration τ also has roots at the periods τ/k : its harmonics

Fundamentals and Harmonics



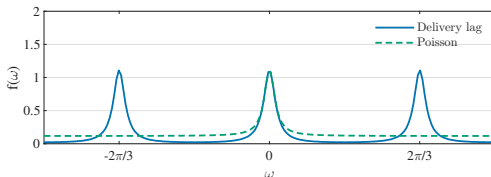
- A cycle with duration $\tau = 6$ and weight $w = 0.75$: autocorrelation w^j at lags $j\tau$, zero elsewhere (bars).
(a) Its Fourier components, darker where the spectrum is larger; black: the fundamental, period τ . (b) Their sum rebuilds the autocorrelations
- The spectrum peaks at the fundamental and at its harmonics, periods τ/k

Poisson Model

- A common way to model delays is **Poisson arrival**: for a delay τ , delivery with probability $1/\tau$ each period
- Example: sector n with a self-loop of weight w . Autocovariance of its output at lag $k \geq 1$:

$$\gamma_k = w \left(1 - \frac{1}{\tau}\right)^{k-1} \frac{1}{\tau} \sigma^2(\hat{y}_{nt}) + \text{further laps}$$

With a delivery lag instead: $\gamma_k = w^j \sigma^2(\hat{y}_{nt})$ at $k = j\tau$, zero at every other lag



Spectrum of a self-loop with $\tau = 3$ and $w = 0.75$: delivery lag vs. Poisson arrival

⇒ Poisson arrival spreads each lap over all later months: the peaks at $2\pi/\tau$ disappear

Delay shocks: a disruption to deliveries

- Echoes do not need a productivity shock: a disruption to deliveries (2021-22) travels the same walks
- The static network has no goods in process and the LP83 pipeline is one month long: neither has room for such a shock
- *Delay shock of T months to n* : at date 0, a fraction ε of the goods due in the next T months will arrive T months late

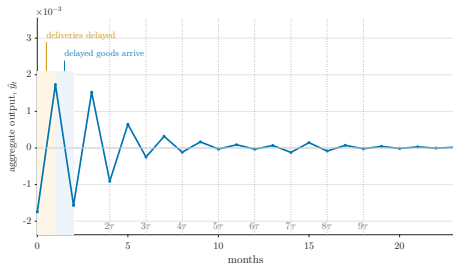
$$\hat{X}_{nj} = -\varepsilon \text{ for } j = 0, \dots, T-1, \quad \hat{X}_{nj} = +\varepsilon \text{ for } j = T, \dots, 2T-1$$

- ▶ X_{nj} : stock of good n that is j months from delivery
 - ▶ T is the length of the disruption; the delay τ_n itself is unchanged
- Nothing is created or destroyed: the goods leave the near end of the pipeline and reappear T months further out

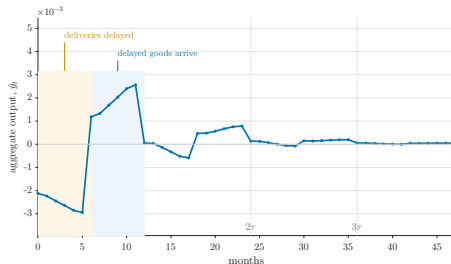
Delay shocks echo at the sector's delay

Response of aggregate output (value-added weighted), $\varepsilon = 1$.

Nonferrous metal, delay 2 months; $T = 1$



Motor vehicle parts, delay 12 months; $T = 6$



- Output falls while the deliveries are missing and rebounds when they arrive; the cumulative response is exactly zero
- The fall and rebound recur at the sector's delay: the echo of the first frame
- Trough: as deep as the impact of a same-size productivity fall (left), 1.6 times as deep (right)

The spectrum of aggregate output: transfer form

Proposition

$$\begin{aligned} f_y(\omega) &= \frac{1}{2\pi} \boldsymbol{\mu}' \mathbf{M}(e^{-i\omega}) \boldsymbol{\Sigma} \mathbf{M}(e^{-i\omega})^* \boldsymbol{\mu} \\ &= \sum_n \mu_n^2 f_n(\omega) + \frac{1}{2\pi} \sum_{n \neq m} \sum_k \mu_n \mu_m [\boldsymbol{\Gamma}_k]_{nm} e^{-i\omega k} \end{aligned}$$

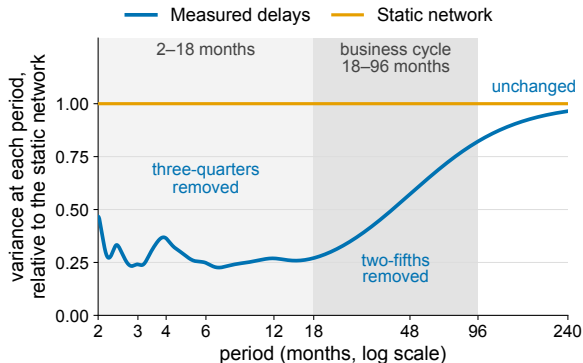
- $\mathbf{M}(z) = (\mathbf{I} - \boldsymbol{\Omega} \text{diag}(z^\tau))^{-1}$: the sum of $w(\varsigma) z^{\tau(\varsigma)}$ over walks, so aggregate output is $\boldsymbol{\mu}' \mathbf{M}(L) \hat{\mathbf{A}}_t$; $\boldsymbol{\Sigma}$: covariance of the shocks; $*$: conjugate transpose
- First line: walks whose arrival dates align at a frequency reinforce each other there; the others offset each other
- Second line: the same quadratic form split into its diagonal terms (sectoral spectra) and its off-diagonal terms (comovements between sectors)

An experiment: same network, same shocks, only the delays differ

- Four economies, same input-output matrix and long-run multiplier
 - ▶ **no delays**: the static network
 - ▶ **one month** everywhere: LP83
 - ▶ **five months** everywhere
 - ▶ the **measured** delays
- Same shocks in all four: persistent, as estimated (sectoral AR(1), common factor)
- Compute the spectrum of aggregate output in each
- Read it **relative to the static network**: $1 =$ the static network's variance at that period

⇒ How much of the network's business-cycle variance do the delays remove?

Delays damp aggregate output at business-cycle periods

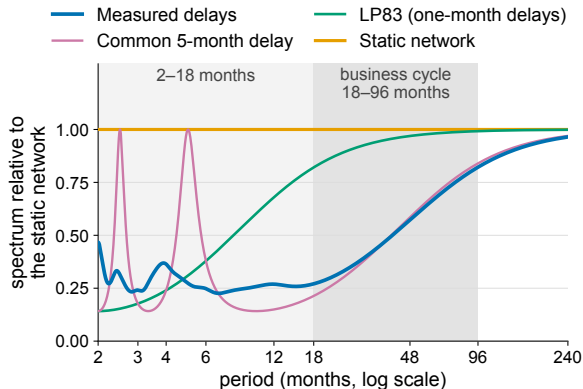


- Long periods: **unchanged**. Each shock has the same total effect, only spread over months
- Business cycle, 18–96 months: **two-fifths** of the variance removed
- 2–18 months: nearly **three-quarters**
- Why: arrivals spread over months partly **cancel** for movements faster than the spread

Overall variance: about 80% of the static network's. Most variance is in slow movements, which the delays leave alone.

⇒ A static reading overstates the network's business-cycle variance by 70%

Dispersed delays damp; one common delay resonates



- Common 5-month delay: every walk arrives at a multiple of 5 months. At the period of 5 months and its harmonic, 2.5, they **line up**: spikes above 1
- Measured delays are **dispersed**: arrivals never line up, only ripples
- LP83: one-month delays, too short to cancel at business-cycle periods

With persistent productivity: our model near one, static at most zero

Why: the predictions 1 and 0 assume i.i.d. growth. Productivity is persistent in the data.

Setup: independent AR(1) productivities, persistence $\rho_n \in [0, 1)$; all delays ≥ 1 month.

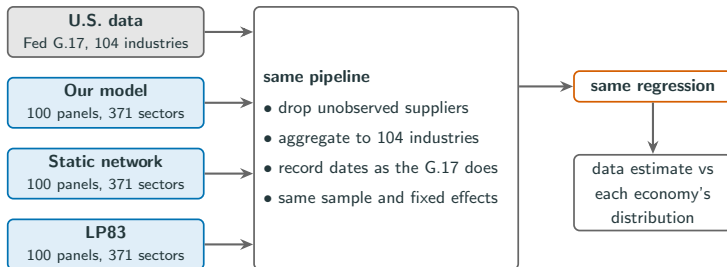
Proposition

Our model: *the slope of $\Delta \hat{y}_{nt}$ on $X_{nt}^{(\tau)}$ is $\beta_\tau = 1 + \text{Cov}(\Delta \hat{A}_{nt}, X_{nt}^{(\tau)}) / \text{Var}(X_{nt}^{(\tau)}) \leq 1$; the gap is of order $(1 - \rho_n)^2$ times the weight of the loops through n .* **Static network:** *the slope is ≤ 0 , on $X^{(\tau)}$ and on any composite with one common lag $k \geq 1$.*

- Why static ≤ 0 : its growth is productivity growth; the composite is past growth; AR(1) growth has nonpositive autocovariance at every lag
- Holds with a common AR(1) factor whose loadings share a sign
- Why growth rates: in levels, persistence alone gives static slopes of 0.8 to 3
- TFP growth is near white noise: annual autocorrelation -0.03 (78 KLEMS series)

Each economy is simulated and measured exactly like the data

- 100 simulated panels per economy, run through the same pipeline as the data: unobserved suppliers dropped, output recorded on the G.17's dates (dating-adjusted design)



- An economy's prediction: the distribution of the estimate across its 100 panels

Two designs for the dates

- **Naive**: treat every series as built-dated. Data 0.33 (0.08); our model 0.99 as the work progresses, 0.14 at delivery
- **Dating-adjusted**: undo the worst case, a sale-dated index running a **full delay** behind production
- Sale-dated **customer** n : its index runs τ_n behind, so its response to a supplier shows up at $\tau_m + \tau_n$. **Shift it back** by τ_n
- Sale-dated **supplier** m : its index runs τ_m behind, so its customers respond to it at lag 0, where the static network also puts it. **Drop it**
- Same shifts and drops in the simulated economies
- It works under either timing: our model predicts 0.62 if sale-dated output is recorded as the work progresses, 0.65 at delivery
- So the panel test holds whatever the timing; the aggregate test is the one that picks it

How recording dates shift measured timing

- An index records output when it is **built** ($\delta = 0$) or at completion or sale ($\delta = \tau$). The measured series is production shifted by δ
- So customer n appears to respond to supplier m at $\tau_m + \delta_n - \delta_m$:

Supplier	Customer	
	built	sale
built	τ_m	$\tau_m + \tau_n$
sale	0	τ_n

- A sale-dated supplier shows at lag 0, where the static network also puts it: **uninformative**
- **Dating-adjusted design:** shift sale-dated customers by τ_n ; weight each supplier by its built-dated share (46 indexes fully, 34 none, 24 partly)
- Same rules applied to the simulated economies

Fed's documentation: 57 built-dated, 31 at completion, 11 at shipment, 4 interpolated, 1 undocumented.

Better-measured delays give a larger coefficient

- **What we do:** rerun the panel test with the **Census-only** delays: measured for the manufacturing core, one imputed value elsewhere

Design	Delays	Data		Simulated economies (mean)		
		$\hat{\beta}_\tau$	(s.e.)	Our model	Static	LP83
Dating-adjusted	Benchmark	0.91	(0.49)	0.65	-0.25	-0.12
Naive	Benchmark	0.33	(0.08)	0.99	0.00	0.06
Dating-adjusted	Census only	0.77	(0.46)	0.68	-0.36	-0.12
Naive	Census only	0.27	(0.06)	1.00	0.00	-0.01

- Replacing imputed delays by measured ones: **0.77 to 0.91**
- Why: a wrong lag misdates a supplier's whole contribution and attenuates the coefficient
- Every row: above all 100 static and all 100 LP83 draws

The aggregate test under both recording readings

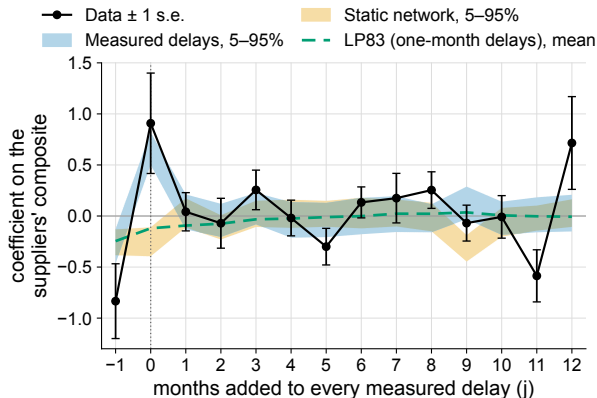
- All 104 industries stay: dropping the sale-dated ones would lose power and change the aggregate
- The simulations **bracket** what a sale date means: **production-timed** (books follow the work, the Census instruction); **worst case** (nothing recorded until delivery)

Simulated economy	$\hat{\beta}_\tau$		Margin over best fixed lag	
	production-timed	worst case	production-timed	worst case
Our model	1.120 ± 0.124	0.116 ± 0.100	0.388 ± 0.116	-0.557 ± 0.112
Static	-0.124 ± 0.185	0.306 ± 0.133	-0.300 ± 0.189	0.005 ± 0.140
LP83	0.158 ± 0.141	0.155 ± 0.141	-1.099 ± 0.170	-1.124 ± 0.154
Data	1.18		+0.91 (excluding lag 12); +0.43 (including)	

- The data match the **production-timed** reading (1.12), not the worst case (0.12). Sample ending in 2019: 1.25, benchmarks unchanged
- Simulated margins exclude the lag-12 cell (no simulated economy has the data's lag-12 structure); the main slide quotes the data's +0.43, which includes it

The comovement peaks at the measured delays

- **What we do:** add j months to every supplier's delay and rerun the dating-adjusted regression, $j = -2, \dots, 12$. Right delays: a **peak at $j = 0$**



- Data: peak at $j = 0$, 2.4 data s.e. above the static band (4.4 in the naive design)
- Also high at $j = +12$, which no economy predicts
- The true delays beat all 500 random reassignments; imputed delays carry nothing; robust to noise

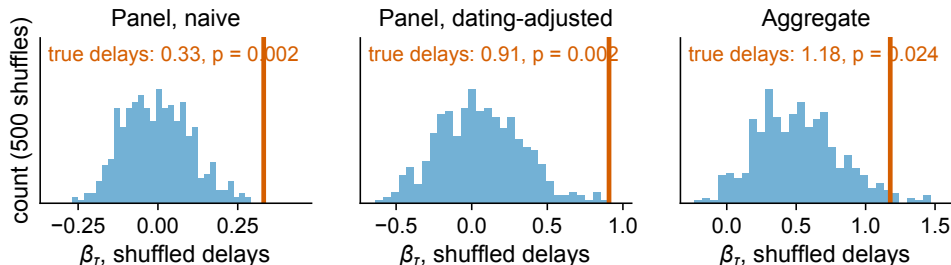
Permutations

Noise

◀ Back

Shuffled delays carry no signal

- **What we do:** reassign the measured delays at random across the 371 sectors, 500 times; rerun the three tests each time; compare with the true delays



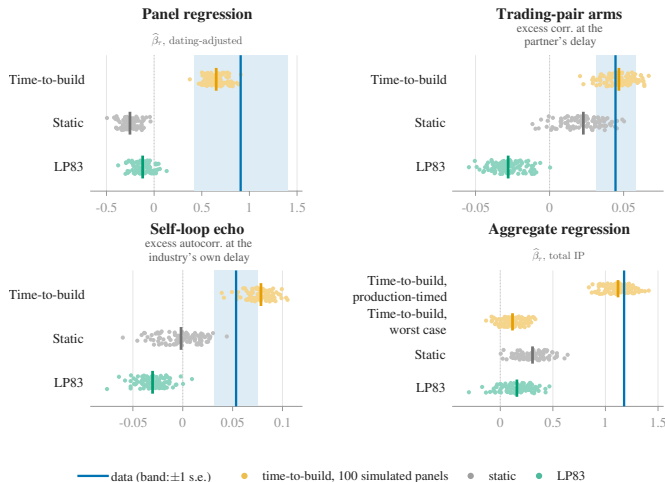
- Panel tests: shuffled coefficients center at **zero**; the true delays beat all 500 ($p = 0.002$)
- Aggregate: shuffled coefficients center at 0.52. Without time effects any dispersed delay vector inherits aggregate momentum. The true delays beat 489 of 500
- So the aggregate identification rests on the **margin over fixed lags**

Measurement noise does not rescue the static network or LP83

- **What we do:** add classical measurement error to every simulated series, rerun the tests
- Calibration: simulated growth variance matches the data's, an extreme reading (about 80% of industry residual variance called noise); scaled by 0.5, 1 and 2
- In every design and at every scale: the data estimate exceeds all 100 static and all 100 LP83 draws
- Noise also lowers our model's prediction. Noise this large would attenuate the data's estimate to 0.1–0.2, below the 0.33–0.91 observed

⇒ The residual variance is better read as non-productivity variation in output

Each test favors our model over the static network and LP83



- Blue line: the estimate in the data, band ± 1 standard error
- Dots: the same estimate on 100 simulated panels per economy; “Time-to-build” is our model
- No parameter is fitted to these data

⇒ Shocks travel through the U.S. network with the measured delays